



Girraween High School

Year 12 HSC Task 2 Examination

Mathematics

February 2016

Time Allowed: 90 Minutes

Instructions:

- There are 12 questions in this paper. All questions are compulsory.
- Start each question (6 – 12) on a new sheet of paper.
- Write on one side of the paper only.
- Show all necessary working.
- Board-approved calculators may be used.
- Marks may be deducted for careless or badly arranged work.

Section I

5 marks

Attempt Questions 1-5

Question 1 (1 mark)

For which values of x is $f(x) = \frac{x^4}{12} - \frac{x^2}{2}$ concave up?

- A. $x > 1$
- B. $x < -1$
- C. $-1 < x < 1$
- D. $x < -1$ and $x > 1$

Question 2 (1 mark)

A function $f(x)$ is increasing at an increasing rate at $x = a$, which of the following is true?

- A. $f'(a) < 0$ and $f''(a) < 0$
- B. $f'(a) < 0$ and $f''(a) > 0$
- C. $f'(a) > 0$ and $f''(a) < 0$
- D. $f'(a) > 0$ and $f''(a) > 0$

Question 3 (1 mark)

Which of the following is the correct value of $\int_0^1 (3x + 1)^3 dx$?

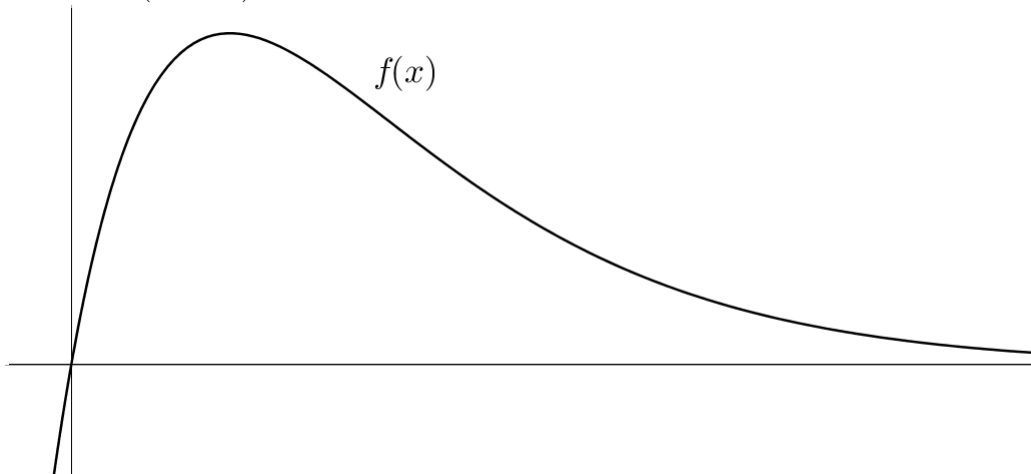
- A. $\frac{64}{3}$
- B. $\frac{85}{4}$
- C. $\frac{255}{4}$
- D. 64

Question 4 (1 mark)

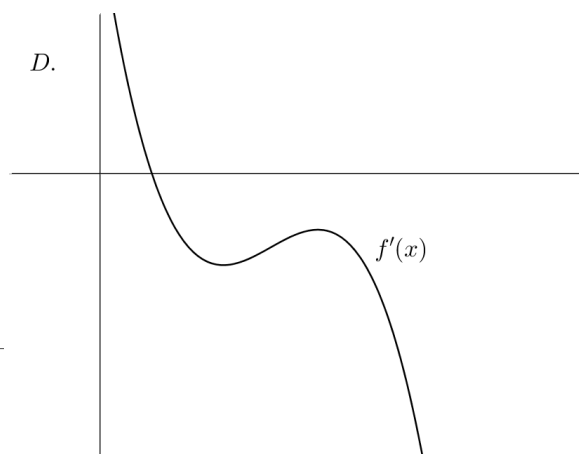
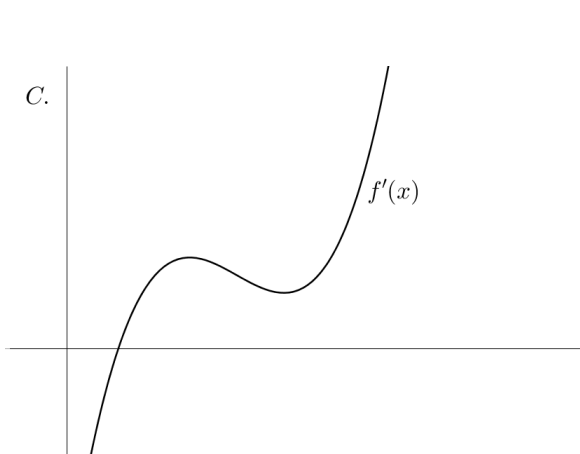
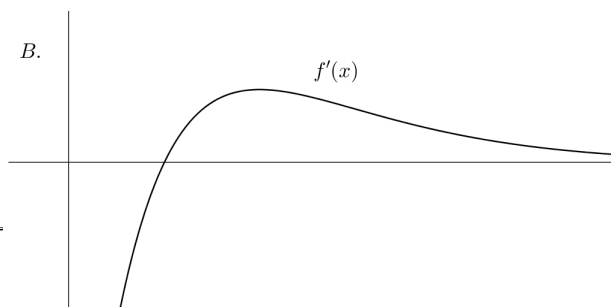
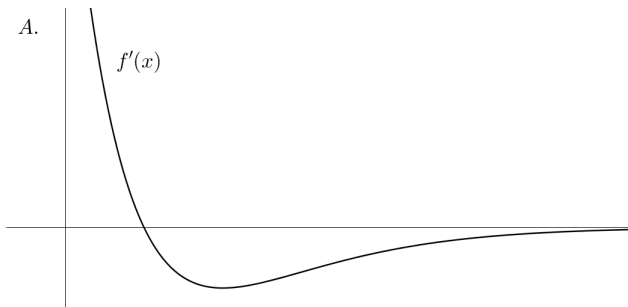
Suppose $b > a$, which of the following is always greater than $\int_a^b f(x) dx$?

- A. $\int_a^b kf(x) dx$ where k is a constant such that $k > 0$
- B. $\int_a^b f(x) + k dx$ where k is a constant such that $k > 0$
- C. $\int_a^b f^2(x) dx$
- D. $\left| \int_a^b f(x) dx \right|$

Question 5 (1 mark)



A graph of $f(x)$ is given above. Which of the following best represents the graph of $f'(x)$?



Section II

71 marks

Attempt Questions 6-12

Write your answers on the paper provided.

In Questions 6-12, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (8 marks)

(a) The curve $y = \sqrt{2ax^2 + x}$ has a stationary point at $x = \frac{1}{2}$. Find the values of a . [3]

(b) If $y = x^2\sqrt{x}$, show that $\frac{3y'}{y''} = 2x$ [3]

(c) Find the values of x such that $y = x(x+1)^3$ is increasing. [2]

Question 7 (12 marks)

(a) Show that the curve $y = \frac{x}{2x+1}$ does not have any points of inflexion. [3]

(b) Consider the function $f(x) = 4x^3 - x^4$.

i. Find the coordinates of the stationary points of the curve $f(x)$, and determine their nature. [4]

ii. Find any points of inflexion. [2]

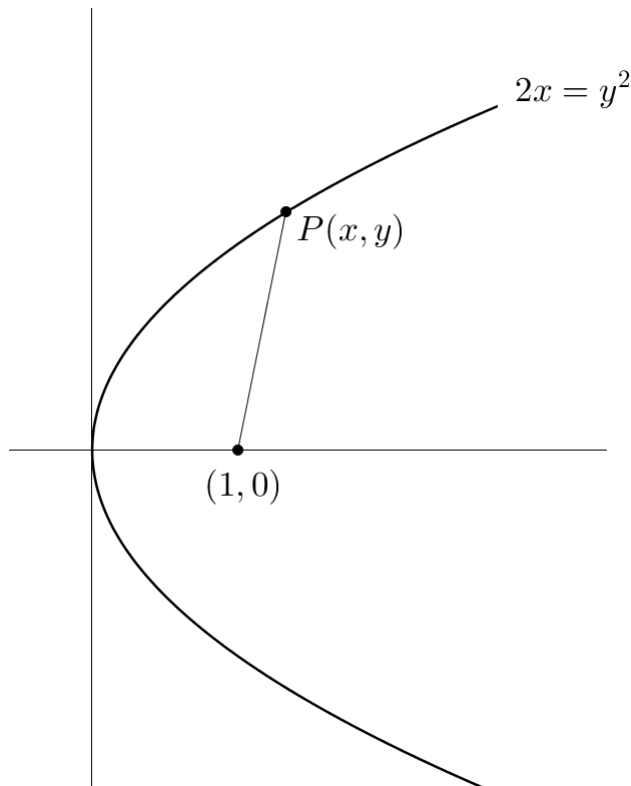
iii. Sketch the graph of $f(x)$, clearly indicating all important features including intercepts. [3]

The exam continues on the next page

Question 8 (8 marks)

- (a) Determine the global minimum value of the function $f(x) = x^4 - 4x^2 + 3$ in the domain $-1 \leq x \leq 2$ [3]

- (b) Suppose point $P(x, y)$ is a point on the parabola $2x = y^2$.



- i. Show that the distance D between $P(x, y)$ and $(1, 0)$ is given by $D = \sqrt{x^2 + 1}$ [2]
- ii. Using calculus, find the coordinates of $P(x, y)$ such that D is minimum. [3]

Question 9 (10 marks)

- (a) Find the primitive of the following:
- i. $f(x) = 2\sqrt{x} - \frac{2}{\sqrt[3]{x}} + 1$ [2]
- ii. $f(x) = \frac{1}{\sqrt{2x+1}}$ [2]
- (b) The gradient function of a curve is given by $y' = 3(x+1)^2$. Find the equation of the curve if it passes through the point $(1, 10)$. [2]
- (c) The second derivative of a curve is given by $\frac{d^2y}{dx^2} = 1 + \frac{1}{x^3}$. Find the equation of the curve given that it passes through the points $(1, 0)$ and $(-1, 5)$. [4]

Question 10 (12 marks)

(a) Find:

i. $\int x^3 + 4x^2 - 1 \, dx$ [1]

ii. $\int 6(1 - 2x)^5 \, dx$ [2]

iii. $\int \sqrt{x} \left(x + \frac{1}{x} \right) \, dx$ [2]

(b) Evaluate:

i. $\int_{-2}^2 x^5 - x^3 + x \, dx$ [2]

ii. $\int_0^{10} \frac{3}{10 \left(\frac{x}{5} - 1 \right)^4} \, dx$ [2]

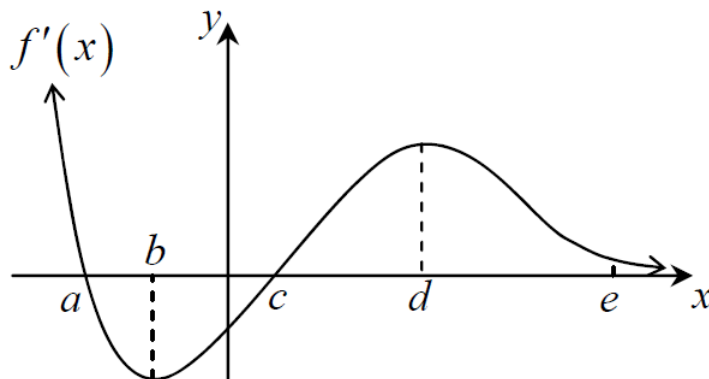
(c) Find the positive value of k such that $\int_1^k k + x \, dx = 0$ [3]

Question 11 (10 marks)

(a) Find the area bounded by the curve $y = x^2 - 4x + 3$ and the x -axis between $x = 2$ and $x = 4$. [3]

(b) Find the area bounded by the curve $y = \sqrt{x - 1}$ and the y -axis between $y = 0$ and $y = 1$ [3]

(c) The graph below represents $y = f'(x)$. Specific x -values a , b , c , d and e are as indicated in the diagram.



i. For what value(s) of x will the graph of $y = f(x)$ have a stationary point? [1]

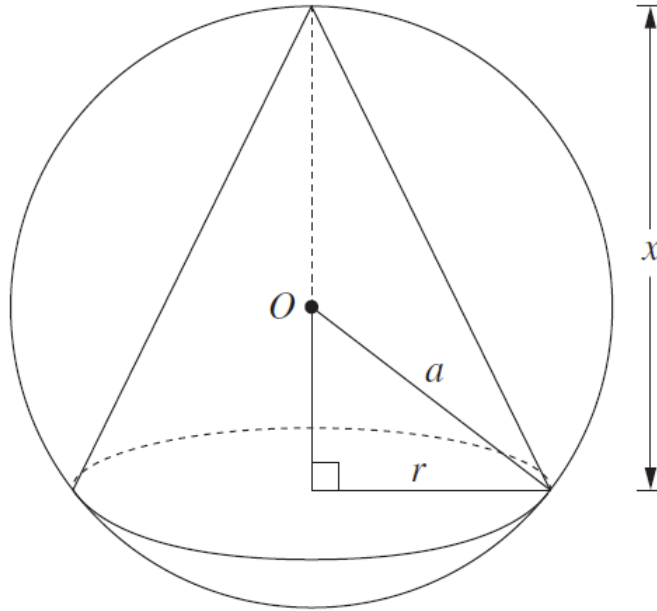
ii. For what value(s) of x is the graph of $y = f(x)$ increasing? [2]

iii. For what value(s) of x is the graph of $f(x)$ concave up? [1]

Question 12 (11 marks)

(a) If $y = ax^3 + bx^2 + cx + d$ has only one stationary point show that $b^2 = 3ac$. [2]

(b) A cone is inscribed in a sphere of radius a , centred at O . The height of the cone is x and the radius of the base is r , as shown in the diagram below.



i. Show that $r^2 = 2ax - x^2$. [2]

ii. Show that the volume, V , of the cone is given by $V = \frac{1}{3}\pi(2ax^2 - x^3)$. [1]

iii. Find the value of x for which the volume of the cone is a maximum. [3]

(c) Find the value of $\int_{-1}^2 |x| + x^2 dx$ [3]

End of exam

Multiple Choice Solutions:

D D B B A

Q6

$$(a) y = \sqrt{2ax^2 + x}$$

$$y = (2ax^2 + x)^{1/2}$$

$$y' = \frac{1}{2} (2ax^2 + x)^{-1/2} \times (4ax + 1)$$

$$y' = \frac{4ax + 1}{2\sqrt{2ax^2 + x}}$$

$$y' = 0 \text{ when } x = \frac{1}{2}$$

$$\therefore 4ax + 1 = 0 \text{ when } x = \frac{1}{2}$$

$$\therefore 2a + 1 = 0 \quad \therefore a = -\frac{1}{2}$$

$$(b) y = x^2 \sqrt{x}$$

$$y = x^{5/2}$$

$$y' = \frac{5}{2} x^{3/2} = \frac{5x\sqrt{x}}{2}$$

$$y'' = \frac{15}{4} x^{1/2} = \frac{15\sqrt{x}}{4}$$

$$\text{Required to prove: } \frac{3y'}{y''} = 2x$$

$$\text{LHS} = \frac{3y'}{y''}$$

$$= \cancel{3} \times \frac{5x\sqrt{x}}{2} \times \frac{4}{15\sqrt{x}}$$

$$= 2x = \text{RHS.}$$

$$(c) y = x(x+1)^3$$

$$u = x \quad v = (x+1)^3$$

$$u' = 1 \quad v' = 3(x+1)^2$$

$$y' = (x+1)^3 + 3x(x+1)^2$$

$$= (x+1)^2 [(x+1) + 3x]$$

$$= (x+1)^2 (4x+1)$$

$$\text{Since } (x+1)^2 \geq 0$$

$$\therefore y' > 0 \text{ when } 4x+1 > 0$$

$$\therefore x > -\frac{1}{4}$$

Q7

$$(a) y = \frac{x}{2x+1}$$

$$u = x \quad v = 2x+1$$

$$u' = 1 \quad v' = 2$$

$$y' = \frac{2x+1-2x}{(2x+1)^2}$$

$$y' = \frac{1}{(2x+1)^2}$$

$$y' = (2x+1)^{-2}$$

$$y'' = -2(2x+1)^{-3} \times 2$$

$$y'' = \frac{-4}{(2x+1)^3} \neq 0 \text{ for all } x.$$

$\therefore$ no points of inflection.

Q7

(b)

$$(i) f(x) = 4x^3 - x^4 = x^3(4-x)$$

$$f'(x) = 12x^2 - 4x^3 = 4x^2(3-x)$$

$$f''(x) = 24x - 12x^2 = 12x(3-x)$$

$$f'(x) = 0 \text{ when } 4x^2(3-x) = 0$$

$$\therefore x = 0 \text{ \& } x = 3$$

$$f(0) = 0 \text{ \& } f(3) = 27$$

$\therefore$ Stationary points are:

$$(0, 0) \text{ \& } (3, 27)$$

$$f''(3) = 12(3)(-1) < 0 \quad \cap$$

$$\therefore (3, 27) \text{ is a max}$$

Since $f''(0) = 0$ using $f'(x)$ table to classify:

x	-1	0	1
$f'(x)$	+	0	+
	/	-	/

$$f'(-1) = 4(4) > 0$$

$$f'(1) = 4(2) > 0$$

$\therefore (0, 0)$ is a horizontal point of inflexion.

(ii)

$$f''(x) = 0 \text{ when } 12x(3-x) = 0$$

$$\therefore x = 0 \text{ \& } x = 3$$

$f(2) = 16 \therefore (2, 16)$ is a possible point of inflexion.

x	1	2	3
$f''(x)$	+	0	-

$$f''(1) = 12(1) > 0$$

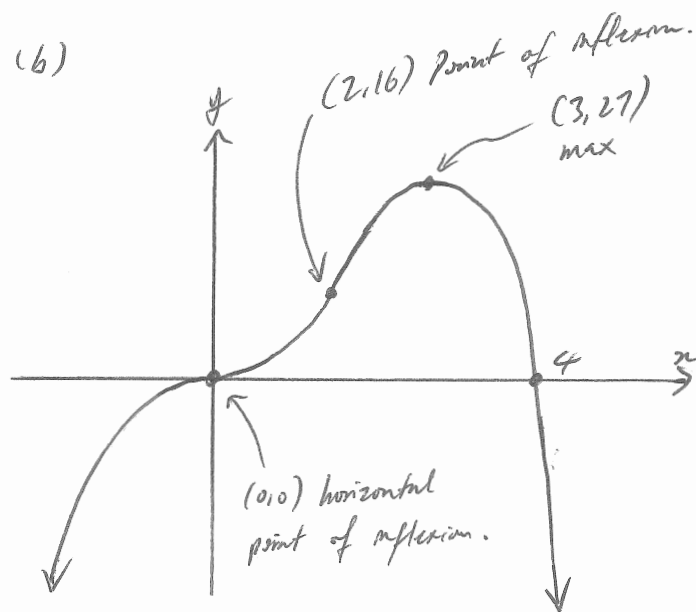
$$f''(3) = 12(3)(-1) < 0$$

$\therefore (2, 16)$ is a point of inflexion

$(0, 0)$ was already classified in

part (i).

(b)



Q3

$$(a) f(x) = x^4 - 4x^2 + 3$$

$$f'(x) = 4x^3 - 8x$$

$$f'(x) = 4x(x^2 - 2)$$

$$f'(x) = 0 \text{ when } 4x(x^2 - 2) = 0$$

$$\text{when } 4x(x - \sqrt{2})(x + \sqrt{2}) = 0$$

$$\therefore x = 0 \text{ \& } x = \pm\sqrt{2}$$

$$f(-1) = 1 - 4 + 3 = 0$$

$$f(2) = 2^4 - 2^4 + 3 = 3$$

$$f(0) = 3$$

$$f(\sqrt{2}) = 4 - 8 + 3 = -1$$

$\therefore$ global minimum value is -1 .

Note: $x = -\sqrt{2}$ is ignored as it is outside of the domain.

(b)

$$(i) D = \sqrt{(x-1)^2 + y^2}$$

$$\text{but } y^2 = 2x$$

$$\therefore D = \sqrt{(x-1)^2 + 2x}$$

$$D = \sqrt{x^2 - 2x + 1 + 2x}$$

$$D = \sqrt{x^2 + 1}$$

(ii)

$$D = (x^2 + 1)^{1/2}$$

$$\frac{dD}{dx} = \frac{1}{2}(x^2 + 1)^{-1/2} \times 2x$$

$$\frac{dD}{dx} = \frac{x}{\sqrt{1+x^2}}$$

$$\therefore \frac{dD}{dx} = 0 \text{ when } x = 0.$$

x	-1	0	1
$\frac{dD}{dx}$	$-$	0	$+$
	$\backslash$	$-$	$/$

$$\left. \frac{dD}{dx} \right|_{x=-1} = -\frac{1}{\sqrt{2}}$$

$$\left. \frac{dD}{dx} \right|_{x=1} = \frac{1}{\sqrt{2}}$$

$\therefore x = 0$ gives minimum D .

$y(0) = 0 \therefore (0, 0)$ is the closest point.

Alternatively:

To minimise D it suffices to minimise D^2

$$D^2 = x^2 + 1$$

$$\frac{dD^2}{dx} = 2x$$

$$\frac{dD^2}{dx} = 0 \text{ when } x = 0$$

$$\frac{d^2D^2}{dx^2} = 2 > 0 \therefore x = 0 \text{ gives minimum } D^2 \text{ which gives minimum } D.$$

Q9

(a)

$$(i) f(x) = 2\sqrt{x} - \frac{2}{\sqrt{x}} + 1$$

$$f(x) = 2x^{\frac{1}{2}} - 2x^{-\frac{1}{2}} + 1$$

$$F(x) = \frac{4}{3}x^{\frac{3}{2}} - 3x^{\frac{1}{2}} + x + C$$

$$F(x) = \frac{4}{3}x\sqrt{x} - 3x^{\frac{1}{2}} + x + C$$

$$(ii) f(x) = \frac{1}{\sqrt{2x+1}}$$

$$f(x) = (2x+1)^{-\frac{1}{2}}$$

$$F(x) = \frac{(2x+1)^{\frac{1}{2}}}{2(\frac{1}{2})} + C$$

$$F(x) = \sqrt{2x+1} + C$$

(b)

$$y' = 3(x+1)^2$$

$$y = 3 \times \frac{(x+1)^3}{3} + C$$

$$y = (x+1)^3 + C$$

$$\text{but } y(1) = 10$$

$$\therefore 10 = 2^3 + C \quad \therefore C = 2$$

$$\therefore y = (x+1)^3 + 2$$

(c)

$$y'' = 1 + \frac{1}{x^3}$$

$$y'' = 1 + x^{-3}$$

$$y' = x - \frac{1}{2}x^{-2} + C$$

$$y = \frac{x^2}{2} + \frac{1}{2}x^{-1} + Cx + D$$

$$y = \frac{1}{2}x^2 + \frac{1}{2x} + Cx + D$$

$$y(1) = 0$$

$$\therefore 0 = \frac{1}{2} + \frac{1}{2} + C + D$$

$$\therefore C + D = -1 \quad \dots (1)$$

$$y(-1) = 5$$

$$\therefore 5 = \frac{1}{2} - \frac{1}{2} - C + D$$

$$\therefore -C + D = 5 \quad \dots (2)$$

$$(1) + (2):$$

$$2D = 4$$

$$\therefore D = 2$$

$$\therefore C = -3$$

$$\therefore y = \frac{1}{2}x^2 + \frac{1}{2x} - 3x + 2$$

Q10

(a)

$$(i) \int x^3 + 4x^2 - 1 \, dx$$

$$= \frac{x^4}{4} + \frac{4}{3}x^3 - x + C$$

$$(ii) \int 6(1-2x)^5 \, dx$$

$$= 6 \int (1-2x)^5 \, dx$$

$$= 6 \times \frac{(1-2x)^6}{-2 \times 6} + C$$

$$= -\frac{1}{2} (1-2x)^6 + C$$

$$(iii) \int \sqrt{x} \left(x + \frac{1}{x}\right) \, dx$$

$$= \int x^{\frac{3}{2}} + x^{-\frac{1}{2}} \, dx$$

$$= \frac{2}{5} x^{\frac{5}{2}} + 2x^{\frac{1}{2}} + C$$

$$= \frac{2}{5} x^2 \sqrt{x} + 2\sqrt{x} + C$$

(b)

$$(i) \int_{-2}^2 (x^5 - x^3 + x) \, dx = 0$$

as $f(x) = x^5 - x^3 + x$ is an odd function.

(b)

$$(ii) \int_0^{10} \frac{3}{10 \left(\frac{x}{5} - 1\right)^4} \, dx$$

$$= \frac{3}{10} \int \left(\frac{x}{5} - 1\right)^{-4} \, dx$$

$$= \frac{3}{10} \times \left[\frac{\left(\frac{x}{5} - 1\right)^{-3}}{-3 \left(\frac{1}{5}\right)} \right]_0^{10}$$

$$= -\frac{1}{2} \left[\frac{1}{\left(\frac{x}{5} - 1\right)^3} \right]_0^{10}$$

$$= -\frac{1}{2} \left[\frac{1}{1} - (-1) \right]$$

$$= -1$$

$$(c) \int_1^k k + x \, dx = 0$$

$$\left[kx + \frac{x^2}{2} \right]_1^k = 0$$

$$k^2 + \frac{k^2}{2} - \left(k + \frac{1}{2}\right) = 0$$

$$k^2 + \frac{k^2}{2} - k - \frac{1}{2} = 0$$

$$2k^2 + k^2 - 2k - 1 = 0$$

$$3k^2 - 2k - 1 = 0$$

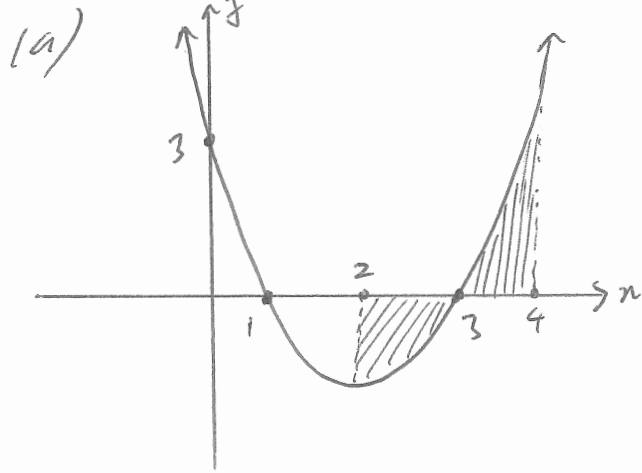
$$\begin{array}{cc} 3k & +1 \\ k & -1 \end{array}$$

$$(3k+1)(k-1) = 0$$

$$\therefore k = -\frac{1}{3} \text{ or } k = 1$$

$$\therefore k = 1 \text{ only as } k > 0$$

Q11) $y = x^2 - 4x + 3 = (x-3)(x-1)$



$$\int_2^3 x^2 - 4x + 3 \, dx$$

$$= \left[\frac{x^3}{3} - 2x^2 + 3x \right]_2^3$$

$$= (3^2 - 2 \times 3^2 + 3^2) - \left(\frac{8}{3} - 8 + 6 \right)$$

$$= -\frac{2}{3}$$

$$\int_3^4 x^2 - 4x + 3 \, dx$$

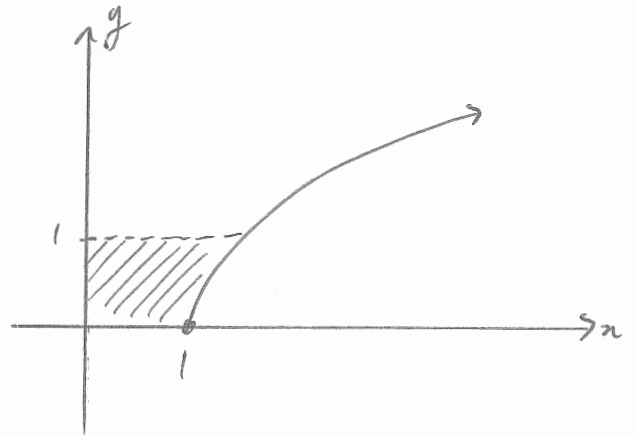
$$= \left[\frac{x^3}{3} - 2x^2 + 3x \right]_3^4$$

$$= \left(\frac{4^3}{3} - 32 + 12 \right) - 0$$

$$= \frac{4}{3}$$

$$\therefore \text{Area} = \frac{2}{3} + \frac{4}{3} = 2 \text{ unit}^2$$

(b)



$$y = \sqrt{x-1}$$

$$y^2 = x-1$$

$$\therefore x = y^2 + 1$$

$$A = \int_0^1 y^2 + 1 \, dy$$

$$= \left[\frac{y^3}{3} + y \right]_0^1$$

$$= \frac{1}{3} + 1 = \frac{4}{3} \text{ unit}^2$$

(c)

(i) $x = a$ & $x = c$

(ii) $x < a$ & $x > c$

(iii) $b < x < d$.

Q12

$$(a) \quad y = ax^3 + bx^2 + cx + d$$

$$y' = 3ax^2 + 2bx + c$$

only one stationary point implies

$3ax^2 + 2bx + c = 0$ only has one solution, which implies

$$\Delta = 0$$

$$\Delta = (2b)^2 - 4(3a)(c) = 0$$

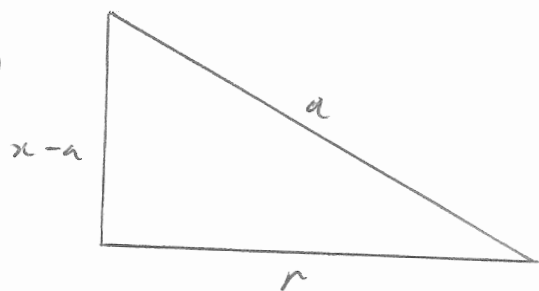
$$4b^2 - 12ac = 0$$

$$b^2 - 3ac = 0$$

$$\therefore b^2 = 3ac$$

(b)

(i)



$$r^2 = a^2 - (x-a)^2$$

$$r^2 = a^2 - (x^2 - 2ax + a^2)$$

$$r^2 = a^2 - x^2 + 2ax - a^2$$

$$r^2 = 2ax - x^2$$

(ii)

$$V = \frac{1}{3} \pi r^2 x$$

$$V = \frac{1}{3} \pi x (2ax - x^2)$$

$$V = \frac{1}{3} \pi (2ax^2 - x^3)$$

(iii)

$$\frac{dV}{dx} = \frac{1}{3} \pi (4ax - 3x^2)$$

$$\frac{d^2V}{dx^2} = \frac{1}{3} \pi (4a - 6x)$$

$$\frac{dV}{dx} = 0 \text{ when } \frac{1}{3} \pi (4ax - 3x^2) = 0$$

$$\frac{1}{3} \pi x (4a - 3x) = 0$$

$$\therefore x = 0 \text{ or } x = \frac{4a}{3}$$

$$\begin{aligned} \left. \frac{d^2V}{dx^2} \right|_{x=\frac{4a}{3}} &= \frac{1}{3} \pi \left[4a - 6\left(\frac{4a}{3}\right) \right] \\ &= \frac{1}{3} \pi (4a - 8a) \\ &= -\frac{4a\pi}{3} \end{aligned}$$

$$\text{Now } a > 0 \therefore -\frac{4a\pi}{3} < 0$$

$$\therefore x = \frac{4a}{3} \text{ gives max } V.$$

$$\text{Note: } \left. \frac{d^2V}{dx^2} \right|_{x=0} = \frac{4a\pi}{3} > 0 \therefore x=0 \text{ gives min } V$$

(c)

$$\text{Since } |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x \leq 0 \end{cases}$$

$$\begin{aligned} \therefore \int_{-1}^2 |x| + x^2 dx &= \int_{-1}^0 -x + x^2 dx + \int_0^2 x + x^2 dx \\ &= \left[-\frac{x^2}{2} + \frac{x^3}{3} \right]_{-1}^0 + \left[\frac{x^2}{2} + \frac{x^3}{3} \right]_0^2 \\ &= 0 - \left(-\frac{1}{2} - \frac{1}{3} \right) + \left(2 + \frac{8}{3} \right) - 0 \\ &= \frac{5}{6} + \frac{14}{3} = \frac{11}{2} \end{aligned}$$